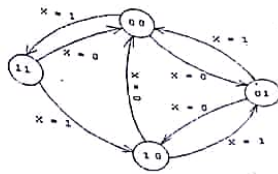
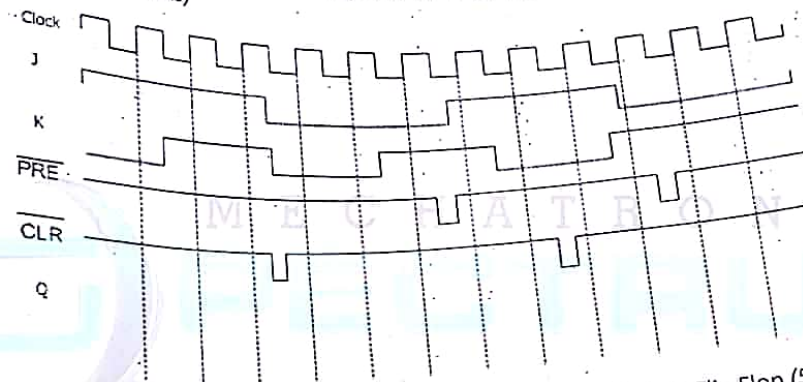


1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	

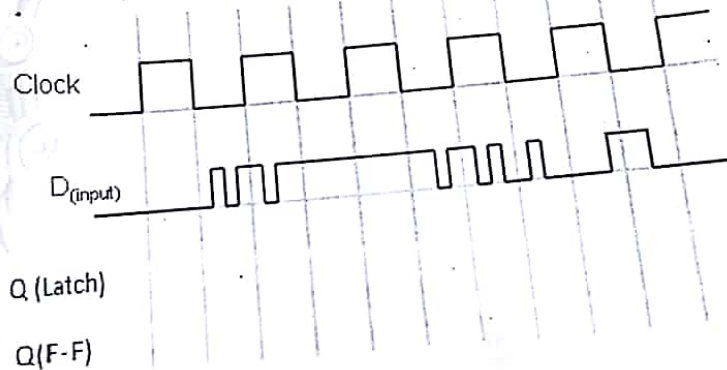
Q2) Draw the circuit which implement the state diagram below using J-K flip flop (10 marks)



Q3)A- For a positive edge triggered J-K flip flop with preset and clear inputs . Determine Q for the inputs shown by the timing diagram. (5 Marks)



B- Complete the following timing diagram for a positive trigger 1- D Latch 2- D Flip Flop (5 Marks)



Q4) Using only Three state buffer(s) and inverter(s), construct a 4-1 MUX with two select lines. (5 Marks)

Q5) Design asynchronous down counter which counts from 15 to 9. (5 Marks)

5) Design a 4 to 2 Priority Decoder such that the inputs are $D_0D_1D_2D_3$, D_0 is the most priority (5 marks)

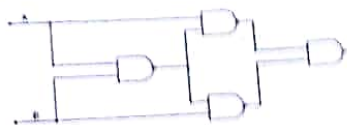
D0	D1	D2	D3	X	Y	V





Q1) Fill in table below the correct answer

- 1- The binary code of $(21.125)_{10}$ is :
A- 10101.001 B- 10100.001 C- 10101.010 D- 10100.111
- 2- One of the following is a universal gate:
A- AND B- NAND C- OR D- XOR
- 3- The number 1000 would appear just immediately after:
A- $FFFF_{16}$ B- 1111_2 C- 7777_8 D- All of above
- 4- Which of the following is a two level logic gate:
A- OR B- AND C- XOR D- NOT gate
- 5- Among the logic families, the family which is used in a systems required high speed operation is:
A- TTL B- CMOS C- MOS D- ECL
- 6- Excess-3 is known as :
A- Weighted code B- Cyclic redundancy code C- Self-complementing code D- Algebraic code
- 7- Indicate which of the following three binary addition is correct:
i. $1011 + 1010 = 10101$ ii. $1010 + 1101 = 10111$ iii. $1010 + 1101 = 11001$
A- i and ii B- i and iii C- iii only D- ii and iii
- 8- Identify the logic function performed by the circuit shown in the given figure



- 9- The Tri-State Buffer is :
A- A combinational circuit B- Sequential circuit C- Memory circuit D- B and C
- 10- What is the minimum number of 2 input NAND gates required to implement the function $(\bar{x} + \bar{y})(z + w)$:
A- 3 B- 4 C- 5 D- 6
- 11- If $A \oplus B = C$, then:
A- $A \oplus C = B$ B- $A \oplus B \oplus C = 0$ C- $B \oplus C = A$ D- A and C
- 12- The circuit shown below is:
A- Full Adder B- Full Subtractor C- Parity Checker D- Multiplier



- 13- Output of the following circuit is:
A- 0 B- 1 C- X D- $\bar{X}$
- 14- The number of 2 input multiplexers required to construct a 2^{10} input multiplexer is:
A- 31 B- 10 C- 127 D- 1023
- 15- Extremely low power dissipation and low cost can be achieved by:
A- MOS B- CMOS C- TTL D- ECL